

C845

Information Systems Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A hospital stores lab results in a Tier III data center with redundant power. The security team publishes a SHA-256 hash alongside each file so clinicians can check that a result has not been altered. Which security goal does publishing the hash support?

- A. Non-repudiation
- B. Confidentiality
- C. Integrity
- D. Availability

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A branch office must send a confidential contract to headquarters using 2048-bit RSA over a VPN the company deployed last year. Only headquarters should be able to read it. Which key should the branch use to encrypt the file?

- A. Headquarters' public key
- B. A shared symmetric key sent in the same email
- C. The branch's private key
- D. Headquarters' private key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A vendor digitally signs a software update. Which key does the customer use to verify the signature?

- A. The customer's private key
- B. The vendor's public key
- C. The customer's public key
- D. The vendor's private key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A credit union's app, which has a 4.8-star rating, requires members to enter a password of at least 14 characters and then answer a security question about their first car. How many authentication factors does this login use?

- A. One factor, because both are something you know.
- B. Strong MFA, because the password is long.
- C. Two factors, because there are two separate prompts.
- D. Two factors: something you know and something you are.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which login combination is true multifactor authentication?

- A. Any two credentials, as long as both are entered
- B. A password plus a PIN
- C. A fingerprint plus a retina scan
- D. A smart card plus a PIN

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A defense contractor with 3,000 employees labels every file Confidential, Secret, or Top Secret. Users receive clearances, and the system grants access only when clearance meets the label; file owners cannot change this. The company recently moved to a new campus. Which access control model is in use?

- A. Role-based access control (RBAC)
- B. Discretionary access control (DAC)
- C. Mandatory access control (MAC)
- D. Rule-based access control

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A retailer's server rack is worth \$200,000 and was installed by a vendor with a five-year warranty. A flood would destroy 25% of its value, and floods are expected once every two years. What is the annualized loss expectancy?

- A. \$50,000
- B. \$25,000
- C. \$200,000
- D. \$100,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A risk has an annualized loss expectancy of \$30,000 and a single loss expectancy of \$10,000. What is its annualized rate of occurrence?

- A. 0.33 times per year
- B. 3 times per year
- C. 10,000 times per year
- D. Cannot be found without the exposure factor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Analysts confirm ransomware on three workstations in the accounting department, which closes its books Friday. The CIO is traveling to a conference. The malware is still trying to reach other hosts. What should the team do next?

- A. Wait for the CIO to return before acting.
- B. Reimage the three workstations right away.
- C. Hold a lessons-learned meeting.
- D. Isolate the infected workstations from the network.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A university's network team, which already runs a SIEM with 90 days of log retention, wants a sensor that sits in the traffic path and automatically drops packets matching known exploit signatures. Which should it deploy?

- A. A firewall rule that only filters by port number
- B. Another SIEM connector
- C. An intrusion prevention system (IPS)
- D. An intrusion detection system (IDS) on a mirror port

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

An attacker exfiltrated data, and the IDS produced no alert. What is this outcome called?

- A. False negative
- B. True negative
- C. True positive
- D. False positive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On a Monday morning, malicious code appeared on a laptop belonging to a sales rep who travels often. Within an hour it had copied itself to 400 other machines by exploiting an unpatched network service, with no one opening files or clicking links. What type of malware is this?

- A. Logic bomb
- B. Virus
- C. Trojan
- D. Worm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A hospital stores lab results in a ~~Tier III data center with redundant power~~. The security team publishes a SHA-256 hash alongside each file so clinicians can check that a result has not been altered. Which security goal does publishing the hash support?

Target word: goal served by the hash **Decides:** A hash detects changes to data.

A	Adjacent category	Needs a digital signature with a private key; a plain hash proves no sender.
B	Confusion set	Hashing does not hide data; encryption does.
C	Correct	Comparing hashes shows whether data was altered.
D	Camouflage	Answers the redundant-power detail, not the hash.

2. Answer A Concept 2

A branch office must send a confidential contract to headquarters using ~~2048-bit RSA over a VPN the company deployed last year~~. Only headquarters should be able to read it. Which key should the branch use to encrypt the file?

Target word: encryption key for confidentiality **Decides:** Only the holder of the matching private key can decrypt.

A	Correct	Only headquarters' private key can decrypt it.
B	Adjacent category	Symmetric keys work, but sending the key with the file destroys secrecy.
C	Confusion set	That creates a signature anyone with the public key can open.
D	Inverted operation	That is the decryption key and should never leave headquarters.

3. Answer B Concept 2 · transfer

A vendor digitally signs a software update. Which key does the customer use to verify the signature?

Target word: verification key **Decides:** Signatures are made with the signer's private key and checked with the signer's public key.

A	Confusion set	Used to decrypt data sent to the customer, not to verify others' signatures.
B	Correct	Anyone can verify with the signer's public key.
C	Adjacent category	Used by others to encrypt to the customer.
D	Inverted operation	That key creates the signature; it is never shared.

4. Answer A Concept 3

A credit union's app, ~~which has a 4.8-star rating~~, requires members to enter a password ~~of at least 14 characters~~ and then answer a security question about their first car. How many authentication factors does this login use?

Target word: count of factor types **Decides:** Password and security question are both something you know.

A	Correct	Factors are counted by category, not by number of prompts.
B	Camouflage	Length adds strength, not factors.
C	Confusion set	Counts steps rather than factor types.
D	Adjacent category	A memory question is not a biometric.

5. Answer D Concept 3 · transfer

Which login combination is true multifactor authentication?

Target word: pick the combo from different categories **Decides:** Different factor categories.

A	Absolute language	Two credentials of the same type are still one factor.
B	Confusion set	Both are something you know.
C	Adjacent category	Both are something you are.
D	Correct	Something you have plus something you know.

6. Answer C Concept 4

A defense contractor ~~with 3,000 employees~~ labels every file Confidential, Secret, or Top Secret. Users receive clearances, and the system grants access only when clearance meets the label; file owners cannot change this. ~~The company recently moved to a new campus.~~ Which access control model is in use?

Target word: name the model **Decides:** System-enforced labels and clearances that owners cannot override.

A	Adjacent category	RBAC ties rights to job roles, not classification labels.
B	Inverted operation	In DAC the owner decides; here owners cannot.
C	Correct	Labels and clearances enforced by the system define MAC.
D	Confusion set	Uses global rules like time of day or IP, not clearances.

7. Answer B Concept 5

A retailer's server rack is worth \$200,000 ~~and was installed by a vendor with a five-year warranty.~~ A flood would destroy 25% of its value, and floods are expected once every two years. What is the annualized loss expectancy?

Target word: ALE **Decides:** $SLE = \$200,000 \times 0.25 = \$50,000$; $ARO = 0.5$.

A	Dropped element	That is the SLE; the yearly rate was not applied.
B	Correct	$\$50,000 \times 0.5 = \$25,000$.
C	Wrong stem number	The asset value.
D	Inverted operation	Divides SLE by 0.5 instead of multiplying.

8. Answer B Concept 5 · transfer

A risk has an annualized loss expectancy of \$30,000 and a single loss expectancy of \$10,000. What is its annualized rate of occurrence?

Target word: solve for ARO **Decides:** $ARO = ALE / SLE$.

A	Inverted operation	Divides SLE by ALE.
B	Correct	$\$30,000 / \$10,000 = 3$.
C	Wrong stem number	Reuses the SLE figure.
D	Adjacent category	EF is only needed to compute SLE, which is given.

9. Answer D Concept 6

Analysts confirm ransomware on three workstations in the accounting department, ~~which closes its books Friday. The CIO is traveling to a conference.~~ The malware is still trying to reach other hosts. What should the team do next?

Target word: next phase after detection **Decides:** The incident is confirmed and still spreading: containment.

A	Camouflage	The response plan should allow containment without waiting.
B	Adjacent category	Eradication before containment lets the malware keep spreading and destroys evidence.
C	Inverted operation	That is the last phase, not the next.
D	Correct	Containment stops the spread before cleanup.

10. Answer C Concept 7

A university's network team, ~~which already runs a SIEM with 90 days of log retention,~~ wants a sensor that sits in the traffic path and automatically drops packets matching known exploit signatures. Which should it deploy?

Target word: device that blocks inline **Decides:** Sits in the path and drops traffic.

A	Dropped element	Lacks signature matching for exploits.
B	Camouflage	A SIEM correlates logs; it does not drop packets.
C	Correct	An inline IPS can block matching traffic.
D	Confusion set	An IDS alerts but does not block.

11. Answer A Concept 7 · transfer

An attacker exfiltrated data, and the IDS produced no alert. What is this outcome called?

Target word: name the missed attack **Decides:** A real attack with no alert.

A	Correct	The system said "no attack" when there was one.
B	Confusion set	No attack and no alert, which is not this case.
C	Adjacent category	An attack that was correctly alerted.
D	Inverted operation	That is an alert when no attack occurred.

12. Answer D Concept 8

~~On a Monday morning,~~ malicious code appeared on a laptop ~~belonging to a sales rep who travels often.~~ Within an hour it had copied itself to 400 other machines by exploiting an unpatched network service, with no one opening files or clicking links. What type of malware is this?

Target word: name the malware type **Decides:** Spread across the network with no user action.

A	Camouflage	The Monday timing hints at a trigger, but self-spreading decides it.
B	Confusion set	A virus needs a host file and user action to spread.
C	Adjacent category	A trojan poses as legitimate software; it does not self-spread.
D	Correct	Worms self-propagate through network vulnerabilities.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CIA triad mapping	Encryption protects confidentiality, hashing protects integrity, and redundancy and backups protect availability.	Thinking a hash keeps data secret.	“Detect changes” vs. “keep secret” vs. “keep running.”
2	Which key does what	Encrypt for confidentiality with the recipient’s public key; sign with the sender’s private key and verify with the sender’s public key.	Encrypting with your own private key for secrecy.	Whose key, and whether the goal is secrecy or proof of origin.
3	Authentication factors	MFA requires factors from different categories: something you know, have, or are.	Counting two passwords or two biometrics as two factors.	Two credentials listed and a question about how many factors.
4	Access control models	DAC = owner decides; MAC = system-enforced labels and clearances; RBAC = permissions attached to job roles.	Calling any centrally managed system MAC.	Who can change the permissions: the owner, the role, or the label.
5	Annualized loss expectancy	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times ARO$.	Stopping at SLE or dividing by ARO.	A dollar value, a percentage of loss, and a yearly frequency.
6	Incident response order	Preparation, detection and analysis, containment, eradication, recovery, lessons learned — contain before you clean.	Wiping or rebuilding systems before the spread is stopped.	“What should the team do next?” after an incident is confirmed.
7	IDS vs. IPS and alert outcomes	An IDS detects and alerts; an inline IPS can block. A false negative is a real attack that raised no alert.	Mixing up false positives and false negatives.	“Block” vs. “alert,” or an attack that was missed.
8	Malware types	A virus attaches to a host and needs user action; a worm self-propagates across networks; a trojan disguises itself as legitimate software.	Calling every spreading infection a virus.	Whether it spread with or without a user doing anything.

Course scope (public description): IT security professionals must be prepared for the operational demands and responsibilities of security practitioners including authentication, security testing, intrusion detection and prevention, incident response and recovery, attacks and countermeasures, cryptography, and malicious code countermeasures. This course provides a comprehensive, up-to-date global body of knowledge that ensures students have the right information, security knowledge, and skills to be successful in IT operational roles to mitigate security concerns and guard against the impact of malicious activity. Students demonstrate how to manage and restrict access control systems; administer policies, procedures, and guidelines that are ethical and compliant with laws and regulations; implement risk management and incident handling processes; execute cryptographic systems to protect data; manage network security; and analyze common attack vectors and countermeasures to assure information integrity and confidentiality in various systems.